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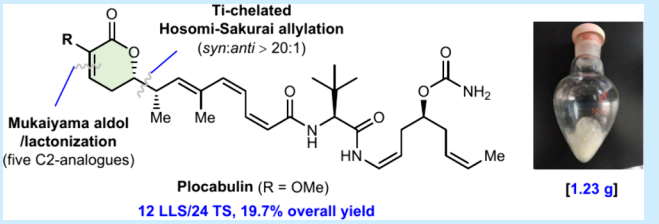
Total Syntheses of Plocabulin and Its C2-Analogues

Shunfa Liu, Bin Li, Xin Gao, Fenglin Wang, Xuan Li, Juxi Wang, Yuzhong Kuang, Luai Ji, Lu Gao, Wanshu Wang, Qifeng Chen,* Yong Qin, and Zhenlei Song*

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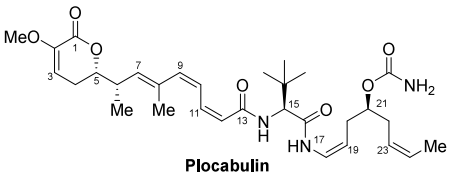
ABSTRACT: Plocabulin is a structurally unique marine natural product with potent antitumor activity, making it a promising candidate for the development of antibody−drug conjugates (ADCs). However, its extremely low natural abundance has limited further research and therapeutic exploration. In this study, we present a convergent synthetic strategy for plocabulin and its C2-analogues. Our approach enabled the synthesis of 1.23 g of plocabulin through a longest linear sequence of 12 steps and a total of 24 steps, achieving an overall yield of 19.7%.



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Plocabulin (PM060184) is a marine-derived natural product isolated from the sponge *Lithoplocamia lithistoides*, collected in Madagascar by Cuevas and co-workers (Scheme 1).[1] It acts as a potent microtubule-destabilizing agent, exhibiting picomolar-level cytotoxicity against cancer cell lines. In certain models, plocabulin is 1,000 to 10,000 times more effective than paclitaxel and vinblastine.[2] It also overcomes P-glycoprotein-mediated multidrug resistance *in vivo*, highlighting its potential in resistant cancer types.[3] Plocabulin binds to a unique site on β-tubulin, partially overlapping with maytansine but distinct from vinblastine, taxanes, and eribulin.[4] This suggests a novel mechanism of action. Currently evaluated in clinical trials for various tumors,[5] plocabulin holds promise for developing next-generation antibody−drug conjugates (ADCs).[6]

Scheme 1. Structures of Plocabulin and Previous Biological, Clinical, Chemical Synthetic and SAR Studies



- **microtubule-destabilizing agent**: GI_{50} = 0.42 nM (HT-29); GI_{50} = 0.59 nM (A-549)
- **clinical trials**: advanced colorectal cancer and breast cancer
- **previous synthesis**: 18 LLS/33 TS, 3.9% yield, 58 mg (Cuevas *et al* 2013)
  18 LLS/34 TS, 4.4% yield, 27 mg (Li/Wang *et al* 2021)
- **SAR studies** on C_2-substitution has not been carried out.

Plocabulin is naturally scarce, making large-scale extraction impractical. PharmaMar isolated only 2.6 mg of plocabulin from 7.66 kg of frozen sponge material, corresponding to a yield of 0.00003%.[1] To date, only two synthetic routes have been reported. Cuevas' approach involves an 18-step longest linear sequence (LLS) and 33 total steps (TS), producing 58 mg of plocabulin with a 3.9% overall yield.[1b] Li and Wang adopted a similar strategy, achieving comparably low efficiency (Scheme 1).[7] Structural studies by Steinmetz revealed a key pharmacophoric interaction between plocabulin's C2-methoxy group and β-tubulin Lys105 through hydrogen bonding.[4] However, synthetic challenges have limited structure−activity relationship (SAR) studies on C2 modifications, despite extensive investigations of other molecular regions.[3,8] Therefore, it is demanding to develop efficient synthetic strategies that not only enable scalable production of plocabulin but also provide access to structurally diverse analogues.

We present a convergent synthesis of plocabulin and its C2 analogues (Scheme 2). A key feature of our strategy is the synthesis of the C1−C8 fragment, which employs a $TiCl_4$-chelated Hosomi−Sakurai allylation to establish the *syn*-C5/C6 stereochemistry, followed by a sequential Mukaiyama aldol-lactonization process to form the α,β-unsaturated lactone with varied C2 substitutions. The efficiency of this route is demonstrated by the gram-scale synthesis of plocabulin (1.23 g) with a 19.7% overall yield in the 12 longest linear sequence (LLS) steps and 24 total steps (TS). This strategy also enabled the synthesis of five C2 analogues, which exhibit antiacute myeloid leukemia (AML) activity comparable to plocabulin. These findings provide valuable insights for the structural optimization of the lactone moiety and the development of improved analogues.

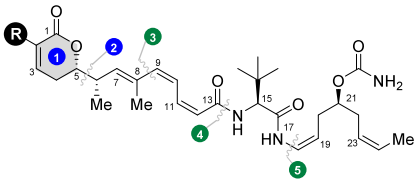
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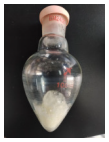


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Scheme 2. Gram-Scale Total Synthesis of Plocabulin and Its C2-Analogues





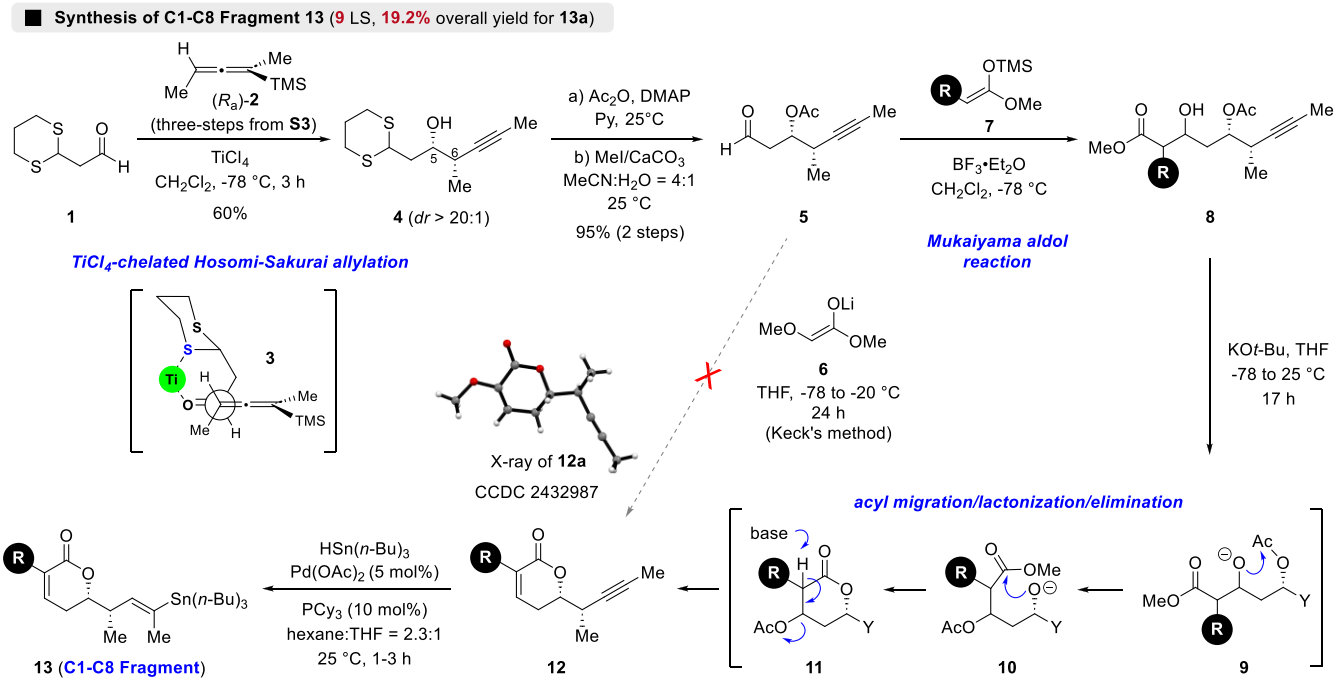
Plocabulin (1.23 g)

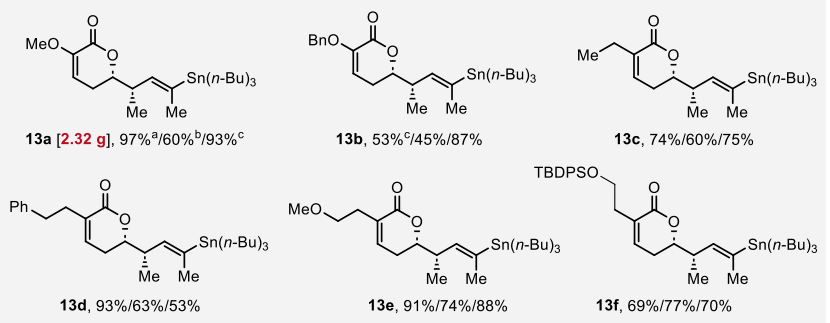
Key steps	**Plocabulin** [R = OMe]: 12 LLS/24 TS/19.7% yield
1 Mukaiyama aldol/lactonization	**PA-1** [R = OBn]: 7.3% yield
2 Hosomi-Sakurai allylation	**PA-2** [R = Et]: 12.5% yield
3 Stille cross-coupling	**PA-3** [R = C_2H_4Ph]: 11.2% yield
4 amidation	**PA-4** [R = $C_2H_4OCH_3$]: 20.5% yield
5 C-N bond cross-coupling	**PA-5** [R = $C_2H_4OCONH_2$]: 13.2% yield

The synthesis of the C1−C8 fragment began with a $TiCl_4$-promoted Hosomi−Sakurai allylation of aldehyde **1**, which contains a 1,3-dithiane at its β-position, using the known chiral allenylsilane (R_a)-**2**[9] (Scheme 3). (R_a)-**2** can be synthesized on 20 g-scale in an overall yield of 62% through a three-step process starting from (*R*)-but-3-yn-2-ol **S3** (see Supporting Information for details). Allylation of α-unsubstituted alkyl aldehydes with allenylsilanes typically results in poor diastereoselectivity, yielding *syn*/*anti* ratios between 2:1 and 4:1.[10] However, in our case, the dithiane moiety in **1** not only served as a masked aldehyde for subsequent transformations but also played a key role in achieving a high *syn*-selectivity. The reaction likely proceeded *via* a Ti(IV)-chelation model (**3**),[11] affording homopropargyl alcohol **4** in 60% yield with a *syn*/*anti* ratio exceeding 20:1. The absolute *S*-configuration at C5 of **4** was confirmed by Mosher ester analysis (see SI for details).[12] Acetylation of **4**, followed by deprotection of the 1,3-dithiane with MeI/$CaCO_3$,[13] provided β-acetoxy aldehyde **5** in 95% yield.

The attempted transformation of β-acetoxy aldehyde **5** *via* Keck's method,[14] employing the lithium enolate of methyl 2-methoxyacetate (**6**) in THF, failed to afford the desired α,β-unsaturated lactone **12a** (Scheme 3). This outcome is consistent with Keck's observation that such transformations are effective only for α-unsubstituted lithium enolates. When

Scheme 3. Synthesis of the C1−C8 Fragment





[a]Yields of Mukaiyama aldol. [b]Yields of lactonization. [c]Yields of hydrostannation.

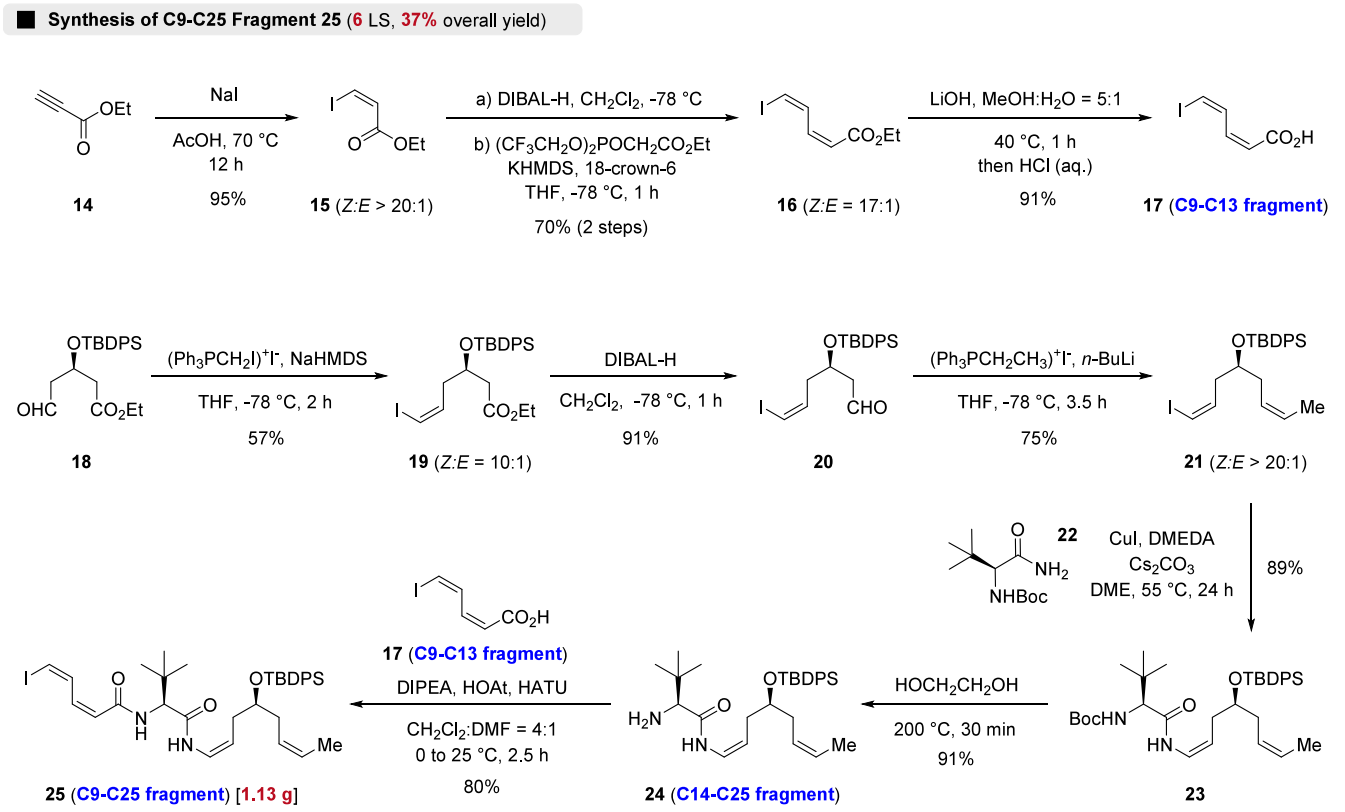
Table 1. Screening reaction conditions (**8a** to **12a**)

entry	base (1.5 eq)	*T* (°C)	yield[a]
1	LDA	-78 to -20	45%
2	LiHMDS	-78 to -20	44%
3	NaHMDS	-78 to -20	47%
4	KHMDS	-78 to -20	52%
5	KO*t*-Bu	-78 to -20	57%
6	KO*t*-Bu	-78 to 0	60%
7	KO*t*-Bu	-78 to 25	65%

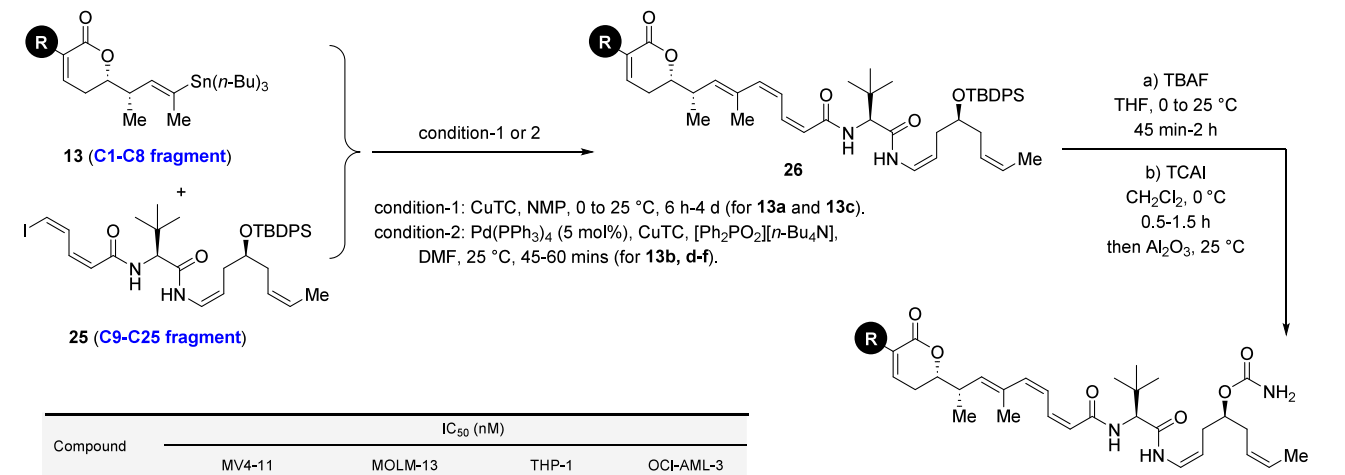
[a]The yields were determined by ^{1}H NMR.

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Scheme 4. Synthesis of the C9−C25 Fragment



Scheme 5. Completion of the Synthesis of Plocabulin and Its C2-Analogues, and the Cytotoxicity of C2-Substituted Derivatives against Four Aggressive Acute Myeloid Leukemia (AML) Cell Lines (MV4-11, MOLM-13, THP-1, and OCI-AML3)



Compound	IC_{50} (nM)			
	MV4-11	MOLM-13	THP-1	OCI-AML-3
Plocabulin	0.0416 ± 0.0042	0.0232 ± 0.0203	0.426 ± 0.148	0.225 ± 0.049
PA-1	0.0424 ± 0.0280	0.0879 ± 0.0122	0.588 ± 0.323	0.366 ± 0.129
PA-2	0.0424 ± 0.0172	0.0299 ± 0.0069	0.890 ± 0.144	0.839 ± 0.244
PA-3	0.0673 ± 0.0321	0.1900 ± 0.0311	0.881 ± 0.208	1.11 ± 0.370
PA-4	0.0777 ± 0.0284	0.0723 ± 0.0230	0.438 ± 0.174	0.423 ± 0.210
PA-5	0.119 ± 0.0172	0.246 ± 0.00681	0.567 ± 0.203	0.436 ± 0.0375

Plocabulin [R = -OMe]: 87%[a] /82%[b]/98%[c] [1.23 g]
PA-1 [R = -OBn]: 64%[d](37%[e])/66%/78%
PA-2 [R = -CH_2CH_3]: 70%/86%/85%
PA-3 [R = -CH_2CH_2Ph]: 81%/85%/68%
PA-4 [R = -$CH_2CH_2OCH_3$]: 81%/90%/97%
PA-5 [R = -$CH_2CH_2OCONH_2$]: 90%/78%/75%

[a]Yields of Stille cross-coupling. [b]Yields of desilylation. [c]Yields of carbamoylation. [d]Yields using condition-2. [e]Yields using condition-1.

the lithium enolate of methyl propionate was used, steric hindrance at the nucleophilic center significantly hindered the aldol reaction. Instead, the enolate acted as a base, triggering proton transfer from the aldehyde methylene, followed by

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elimination. To address this limitation, we developed a two-step approach. First, a $BF_3{\cdot}OEt_2$-promoted Mukaiyama aldol reaction[15] between **5** and silyl enol ether **7a** produced **8a** in 97% yield. In the subsequent lactonization step, KO*t*-Bu was the most effective base among those tested, enabling the formation of lactone **12a**[16] in 60% yield *via* sequential acyl migration, lactonization, and elimination (**9** to **11**). This strategy was broadly applicable, allowing the synthesis of lactones **12** with diverse C2 substituents, including OBn, Et, and alkyl chains. Finally, *cis*-hydrostannation of the alkyne converted **12** into *E*-vinyltributylstannane **13**.[17] Using this streamlined approach, the C1−C8 fragment of plocabulin (**13a**, 2.3 g) was synthesized over 9 linear steps, achieving an overall yield of 19.2%.

The C9−C25 fragment was constructed by coupling dienyl carboxylic acid **17** (C9−C13) with amine **24** (C14−C25) *via* C13−N14 amide bond formation. The C9−C13 fragment was synthesized from ethyl propiolate (**14**) following Nicolaou's protocol (Scheme 4).[18] Treatment of **14** with NaI in AcOH yielded vinyl iodide **15** in 95% yield with a >20:1 *Z*/*E* ratio. Ester reduction generated a labile aldehyde, which immediately underwent a Still-Gennari reaction at − 78 °C, affording dienyl ester **16** in 70% overall yield with a 17:1 *Z*/*E* ratio. Saponification with LiOH in $MeOH/H_2O$ provided carboxylic acid **17** in 91% yield, which was used directly without purification.

The C14−C25 fragment synthesis began with aldehyde **18**, either commercially available or accessible *via* a known three-step route (Scheme 4).[19] A sequential Wittig olefination strategy was employed, using nonstabilized ylides in THF—a dipolar aprotic solvent. The conditions favor formation of the *cis*-oxaphosphetane intermediate under kinetic control, resulting in high *Z*-selectivity at both the C18−C19 and C23−C24 alkene positions.[20] The first Wittig reaction of aldehydes **18** with (iodomethyl)triphenylphosphonium iodide furnished **19** in 57% yield with a *Z*/*E* ratio of 10:1.[21] Subsequent DIBAL-H reduction followed by a second Wittig reaction with ethyltriphenylphosphonium iodide afforded **21** in 75% yield with >20:1 *Z*/*E* selectivity. This three-step sequential Wittig strategy significantly improved efficiency compared to Cuevas' eight-step method.[1b] Cross-coupling of **21** with primary amide **22** using copper-catalysis developed by Li and Wang[7] gave **23** in 89% yield. Boc-deprotection of **23** produced primary amine **24**, which was then condensed with C9−C13 fragment **17**, affording amide **25** in 80% yield. Overall, the C9−C25 fragment of plocabulin (**25**, 1.1 g) was synthesized in 6 linear steps with an overall yield of 37%. The final assembly of plocabulin was achieved through a CuTC-mediated Stille cross-coupling[22] (condition-1) between **13a** and **25**, giving **26a** in 87% yield on a near-gram-scale (911.6 mg prepared; Scheme 5). Desilylation of **26a** produced alcohol **27a**, which underwent carbamoylation with trichloroacetyl isocyanate (TCAI) to afford plocabulin in a 98% yield. The highly efficient final three-step sequence enabled us to obtain 1.23 g of plocabulin, which is sufficient to meet the clinical trial requirements for approximately 75 cancer patients based on current dosing protocols.[5] In total, our synthesis of plocabulin was completed in 12 longest linear sequence (LLS) steps and 24 total steps (TS) with an overall yield of 19.7%. The analytical data for our synthetic plocabulin, including ^1H and ^{13}C NMR spectroscopy, high-resolution mass spectrometry, and optical rotation, closely matched those reported for the natural product and previously synthesized samples.[1b, 7] The final three-step transformation was also applied to the synthesis of plocabulin C2-analogues (**PA-1** to **PA-5**). Notably, variations in the C2 substituent significantly influenced the efficiency of the Stille cross-coupling reaction. For example, CuTC-mediated coupling of C2-OBn-substituted **13b** with **25** resulted in substantial recovery of unreacted starting material, yielding **26b** in only 37% even after an extended reaction time of 4 days (Scheme 5). Drawing inspiration from Fürstner's work,[23] we adopted $Pd(PPh_3)_4$ catalysis (condition-2), which significantly improved the yield of **26b** to 64%.

The lactone ring is a well-established pharmacophore essential for plocabulin's antitumor activity. However, the impact of structural modifications at the C2 position remains largely unexplored. To address this, we assessed the cytotoxicity of C2-substituted derivatives against four aggressive acute myeloid leukemia (AML) cell lines (MV4-11, MOLM-13, THP-1, and OCI-AML3). As shown in Scheme 5, all derivatives exhibited antiproliferative activity comparable to plocabulin, maintaining significant potency at the picomolar level. Notably, there was no substantial difference in activity between derivatives with hydrophobic substituents (benzyloxy, ethyl, and phenethyl; **PA-1**, **PA-2**, and **PA-3**) and those with terminal hydrophilic groups (methoxy and carbamate; **PA-4**, and **PA-5**). These findings indicate that C2 modifications have minimal impact on plocabulin's antitumor activity, offering valuable insight for future structural simplification and optimization efforts.

In summary, we have developed a concise gram-scale synthesis of plocabulin. Our approach efficiently assembles the C1−C8, C9−C13, and C14−C25 fragments in a highly convergent manner. Key steps include a $TiCl_4$-chelated Hosomi−Sakurai allylation to install the *syn*-C5/C6 stereochemistry, followed by a sequential Mukaiyama aldol/lactonization process to construct the α,β-unsaturated lactone core with diverse C2 substitutions. Our approach not only addresses the supply challenges of plocabulin, supporting the current clinical studies and exploration of new therapeutic indications, but also offers a promising platform for developing improved analogues with enhanced therapeutic potential.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.5c01284.

Experimental procedures, spectra data, and crystallographic data (**12a**) for products (PDF)

Accession Codes

Deposition Number 2432987 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge *via* the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

■ AUTHOR INFORMATION

Corresponding Authors

Qifeng Chen − *Key Laboratory of Drug-Targeting and Drug Delivery System of the Education Ministry and Sichuan Province, Sichuan Engineering Laboratory for Plant-Sourced*

https://doi.org/10.1021/acs.orglett.5c01284
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Drug and Sichuan Research Center for Drug Precision Industrial Technology, West China School of Pharmacy, Sichuan University, Chengdu 610041, China; Email: qfchen@scu.edu.cn

Zhenlei Song − *Key Laboratory of Drug-Targeting and Drug Delivery System of the Education Ministry and Sichuan Province, Sichuan Engineering Laboratory for Plant-Sourced Drug and Sichuan Research Center for Drug Precision Industrial Technology, West China School of Pharmacy, Sichuan University, Chengdu 610041, China*; orcid.org/0000-0002-7228-1572; Email: zhenleisong@scu.edu.cn

Authors

Shunfa Liu − *Key Laboratory of Drug-Targeting and Drug Delivery System of the Education Ministry and Sichuan Province, Sichuan Engineering Laboratory for Plant-Sourced Drug and Sichuan Research Center for Drug Precision Industrial Technology, West China School of Pharmacy, Sichuan University, Chengdu 610041, China*

Bin Li − *Key Laboratory of Drug-Targeting and Drug Delivery System of the Education Ministry and Sichuan Province, Sichuan Engineering Laboratory for Plant-Sourced Drug and Sichuan Research Center for Drug Precision Industrial Technology, West China School of Pharmacy, Sichuan University, Chengdu 610041, China*

Xin Gao − *Key Laboratory of Drug-Targeting and Drug Delivery System of the Education Ministry and Sichuan Province, Sichuan Engineering Laboratory for Plant-Sourced Drug and Sichuan Research Center for Drug Precision Industrial Technology, West China School of Pharmacy, Sichuan University, Chengdu 610041, China*

Fenglin Wang − *Key Laboratory of Drug-Targeting and Drug Delivery System of the Education Ministry and Sichuan Province, Sichuan Engineering Laboratory for Plant-Sourced Drug and Sichuan Research Center for Drug Precision Industrial Technology, West China School of Pharmacy, Sichuan University, Chengdu 610041, China*

Xuan Li − *Key Laboratory of Drug-Targeting and Drug Delivery System of the Education Ministry and Sichuan Province, Sichuan Engineering Laboratory for Plant-Sourced Drug and Sichuan Research Center for Drug Precision Industrial Technology, West China School of Pharmacy, Sichuan University, Chengdu 610041, China*

Juxi Wang − *Key Laboratory of Drug-Targeting and Drug Delivery System of the Education Ministry and Sichuan Province, Sichuan Engineering Laboratory for Plant-Sourced Drug and Sichuan Research Center for Drug Precision Industrial Technology, West China School of Pharmacy, Sichuan University, Chengdu 610041, China*

Yuzhong Kuang − *Key Laboratory of Drug-Targeting and Drug Delivery System of the Education Ministry and Sichuan Province, Sichuan Engineering Laboratory for Plant-Sourced Drug and Sichuan Research Center for Drug Precision Industrial Technology, West China School of Pharmacy, Sichuan University, Chengdu 610041, China*

Luai Ji − *Key Laboratory of Drug-Targeting and Drug Delivery System of the Education Ministry and Sichuan Province, Sichuan Engineering Laboratory for Plant-Sourced Drug and Sichuan Research Center for Drug Precision Industrial Technology, West China School of Pharmacy, Sichuan University, Chengdu 610041, China*

Lu Gao − *Key Laboratory of Drug-Targeting and Drug Delivery System of the Education Ministry and Sichuan Province, Sichuan Engineering Laboratory for Plant-Sourced Drug and Sichuan Research Center for Drug Precision Industrial Technology, West China School of Pharmacy, Sichuan University, Chengdu 610041, China*

Wanshu Wang − *Key Laboratory of Drug-Targeting and Drug Delivery System of the Education Ministry and Sichuan Province, Sichuan Engineering Laboratory for Plant-Sourced Drug and Sichuan Research Center for Drug Precision Industrial Technology, West China School of Pharmacy, Sichuan University, Chengdu 610041, China*

Yong Qin − *Key Laboratory of Drug-Targeting and Drug Delivery System of the Education Ministry and Sichuan Province, Sichuan Engineering Laboratory for Plant-Sourced Drug and Sichuan Research Center for Drug Precision Industrial Technology, West China School of Pharmacy, Sichuan University, Chengdu 610041, China*; orcid.org/0000-0003-3434-5747

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.orglett.5c01284

Notes

The authors declare no competing financial interest.

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(5) (a) The data of clinical trials for Plocabulin is available under accession number NCT01299636, NCT02533674, NCT03427268, and can be obtained free of charge from the National Center for Biotechnology Information *via* 2024 https://clinicaltrials.gov/. (b) The data of clinical trials for Plocabulin is available under accession

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